

# Di Hou

National University of Singapore

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## Appointments

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| <b>Research Fellow (Postdoctoral)</b>                       | 2026 – 2027 (expected) |
| Department of Mathematics, National University of Singapore | Singapore              |
| Advisor: Prof. Kim-Chuan Toh                                |                        |

## Education

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| <b>Ph.D. in Mathematics   Advisor: Prof. Kim-Chuan Toh</b>   | 2022 - 2026      |
| Department of Mathematics, National University of Singapore, | Singapore        |
| <b>Bachelor of Science in Mathematics (Honors)</b>           | 2018 - 2022      |
| Department of Mathematics, Jilin University,                 | Changchun, China |

## Research Interests

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- Mathematical Optimization
- Scientific Computing

## Publications

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- [1] [Di Hou](#), Tianyun Tang, Kim-Chuan Toh, Shiwei Wang. On the efficient computation of proximal operators of affine-constrained nonconvex functions.  
**Mathematics of Operations Research**, accepted. [arXiv]
- [2] [Di Hou](#), Tianyun Tang, Kim-Chuan Toh. A low-rank augmented Lagrangian method for Polyhedral-SDP and Moment-SOS relaxations of polynomial optimization.  
**Mathematical Programming** (2026), pp. 1–53. [DOI]
- [3] [Di Hou](#), Tianyun Tang, Kim-Chuan Toh. RiNNAL+: a Riemannian ALM solver for SDP-RLT relaxations of mixed-binary quadratic programs.  
**Mathematical Programming Computation**, 18 (2026), pp. 679–735. [DOI]
- [4] [Di Hou](#), Tianyun Tang, Kim-Chuan Toh. A low-rank augmented Lagrangian method for doubly nonnegative relaxations of mixed-binary quadratic programs.  
**Operations Research**, 74(3) (2025), pp. 1626–1647. [DOI]
- [5] [Di Hou](#), Ling Liang, Kim-Chuan Toh. A sparse smoothing Newton method for solving discrete optimal transport problems.  
**ACM Transactions on Mathematical Software**, 50(3) (2024), pp. 1–26. [DOI]
- [6] Jinda Yang, Haiming Song, Xinxin Li, [Di Hou](#). Block mirror stochastic gradient method for stochastic optimization.  
**Journal of Scientific Computing**, 94, Article 69 (2023). [DOI]

## Preprints

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\* Corresponding author.

- [1] Di Hou, Kim-Chuan Toh. A logarithmic-dual-update Bregman ADMM for optimal transport. arXiv:2610.08087, 2026.
- [2] Kristo Nugraha Lian, Nehal Ahmed Shaikh, Di Hou<sup>\*</sup>, Xingyu Xie, Kim-Chuan Toh. McADMM: A Multi-Clique Augmented Lagrangian-Based Algorithm for Large-Scale Sparse SDPs with Bound Constraints. **ACM Transactions on Mathematical Software**, submitted. [arXiv]
- [3] Lei Yang, Haihang Lan, Di Hou<sup>\*</sup>, Ling Liang, Kim-Chuan Toh. An inexact Halpern-accelerated preconditioned generalized proximal point algorithm for the maximal monotone inclusion problem. **Journal of Scientific Computing**, submitted. [arXiv]
- [4] Di Hou, Kim-Chuan Toh. A dual-certificate procedure for cut generation and presolving in polynomial optimization. **SIAM Journal on Optimization**, submitted.
- [5] Nachuan Xiao, Di Hou<sup>\*</sup>, Kim-Chuan Toh. Dual parameterization for nonconvex constrained optimization. **SIAM Journal on Optimization**, submitted.
- [6] Di Hou, Thai P.D. Nguyen, Kim-Chuan Toh, Guanyi Wang. Sparsity-cone SDP relaxations and applications to variable fixing for sparse quadratic programs. **Operations Research**, submitted. [arXiv]

## Research Software

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- **RiNNAL+**: MATLAB solver for SDP–RLT relaxations of mixed-binary quadratic programs. Jointly developed with Tianyun Tang and Kim-Chuan Toh. [Code]
- **RiNNAL**: MATLAB solver for doubly nonnegative relaxations of mixed-binary quadratic programs. Jointly developed with Tianyun Tang and Kim-Chuan Toh. [Code]

## Recognitions

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- Top Graduate Researcher Award, Faculty of Science, NUS, 2026.
- Best Graduate Researcher Award, Department of Mathematics, NUS, 2026.
- EASIAM 2026 Travel Award, 2026.
- PhD Conference Award, NUS, 2025.
- President’s Graduate Fellowship, NUS, Singapore, 2022-2026.
- Graduate Tutor Merit Award (Tier 1), NUS, Singapore, 2024, 2025, 2026.
- Top Ten Undergraduate Student Award, Changchun City, China, 2022.
- Top Ten Undergraduate Student Award, Jilin University, China, 2022.
- National First Prize Award, Chinese Mathematics Competitions, 2021.
- National Scholarship, China, 2018-2022.

## Teaching

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- MA 4260 Stochastic Operations Research, semester 2, AY 2025/26
- DSA2102 Essential Data Analytics Tools: Numerical Computation, semester 2, AY 2024/25
- MA 3236 Non-Linear Programming, semester 1, AY 2024/25
- DSA3102 Essential Data Analytics Tools: Convex Optimisation, semester 2, AY 2023/2024
- DSA3102 Essential Data Analytics Tools: Convex Optimisation, semester 1, AY 2023/2024

## Talks

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### **RiNNAL-X: Low-rank ALM for structured SDP relaxations**

Computational Robotics Group Seminar, Harvard University (online) June 2026

### **RiNNAL-POP: A low-rank augmented Lagrangian method for polyhedral-SDP and moment-SOS relaxations of polynomial optimization**

19th East Asia SIAM (EASIAM) Annual Meeting, Jeju, Korea Aug. 2026

SIAM Conference on Optimization (OP26), Edinburgh, UK June 2026

Mini Workshop on Optimization, NUS Dec. 2025

### **RiNNAL+: A Riemannian ALM solver for SDP-RLT relaxations of MBQP**

8th PKU Workshop for Outstanding PhD Students in Computational and Applied Mathematics, Peking University, Beijing, China Nov. 2025

International Conference on Continuous Optimization (ICCOPT), Los Angeles, U.S. July 2025

### **RiNNAL: A low-rank ALM for doubly nonnegative relaxations of mixed-binary QP**

9th NUS Graduate Symposium in Mathematics, NUS May 2025

### **A sparse smoothing Newton method for solving discrete optimal transport problems**

NUS-SJTU PhD Forum, NUS Nov. 2024

## Service

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### **Referee for Journals**

- Optimization Methods and Software
- ACM Transactions on Mathematical Software
- Optimization and Engineering
- Numerical Algebra, Control and Optimization

### **Other Services**

- President of NUS SIAM Student Chapter, Jul. 2025 – Jul. 2026.
- Secretary of NUS SIAM Student Chapter, Jul. 2024 – Jul. 2025.
- Vice president of NUS Mathematics Graduate Society, Jul. 2025 – Present.